# CurriculumAI student evaluation analysis

*Descriptive results package for RQ4*

**Design and Evaluation of a Retrieval-Augmented Question Answering System for Computer Programming Students. BSc Computer Science, De Montfort University.**

## 1 Dataset overview and completeness

The supplied Microsoft Forms export contains seven questionnaire submissions in Sheet1, with 38 columns. Section 2 (I–W) concerns CurriculumAI and Section 3 (X–AL) concerns ChatGPT; both system sections belong to the same respondent within each row. Non-identifying labels R01–R07 follow source rows 2–8. All seven respondents reported previous Scala study. Four were assigned CurriculumAI first and three ChatGPT first; this field is contextual and does not determine section ownership.

The only missing rating is P8, CurriculumAI SUS item 8 for R07. Item-level results exclude that cell and use six valid ratings for that item. Every other rating item uses seven. No missing value is imputed. Personal metadata are excluded from the results package.

## 2 SUS results using available complete scores

SUS scoring follows the standard Brooke (1996) method: odd-item ratings minus 1 and 5 minus even-item ratings are summed and multiplied by 2.5. A total requires all ten items. Scores are 0–100 points, not percentages of agreement. The standard deviation is the sample SD.

| System | n | Mean | Median | Minimum | Maximum | Sample SD |
|---|---|---|---|---|---|---|
| CurriculumAI | 6 | 80.83 | 78.75 | 70.00 | 100.00 | 10.80 |
| ChatGPT | 7 | 60.71 | 57.50 | 42.50 | 90.00 | 14.70 |

CurriculumAI has six complete SUS scores. ChatGPT has seven, including R07’s score of 90.00. These unequal available-data samples are reported for completeness; the primary within-participant comparison uses the same six complete submissions for both systems.

## 3 Matched pair SUS comparison

The matched analysis includes R01–R06 only. CurriculumAI had a mean SUS score of 80.83 and ChatGPT 55.83. The mean within-participant difference was 25.00 points and the median difference was 23.75 points. All six differences were positive; none were negative or zero. These are descriptive observations. No statistical significance test was performed.

| System | n | Mean | Median | Minimum | Maximum | Sample SD |
|---|---|---|---|---|---|---|
| CurriculumAI | 6 | 80.83 | 78.75 | 70.00 | 100.00 | 10.80 |
| ChatGPT | 6 | 55.83 | 57.50 | 42.50 | 65.00 | 7.69 |

| Record | CurriculumAI | ChatGPT | Paired difference | Pair status |
|---|---|---|---|---|
| R01 | 80.00 | 57.50 | 22.50 | Included |
| R02 | 72.50 | 60.00 | 12.50 | Included |
| R03 | 85.00 | 57.50 | 27.50 | Included |
| R04 | 100.00 | 42.50 | 57.50 | Included |
| R05 | 77.50 | 52.50 | 25.00 | Included |
| R06 | 70.00 | 65.00 | 5.00 | Included |
| R07 | Not calculated | 90.00 | Not calculated | Excluded: missing CurriculumAI SUS8 (P8) |

Individual differences ranged from 5.00 to 57.50 points. R07's CurriculumAI total and paired difference are not calculated because SUS8 is missing. Its ChatGPT score remains in the seven-score available-data description. That unpaired score cannot establish R07's relative preference between systems.

## 4 Perceived usefulness

U1: This system provided explanations that were useful for understanding Scala concepts.

U2: This system helped me complete the assigned programming question-answering tasks.

| Item | System | n | Median | Disagreement 1–2 | Neutral 3 | Agreement 4–5 |
|---|---|---|---|---|---|---|
| U1 | CurriculumAI | 7 | 4 | 0/7 (0.0%) | 2/7 (28.6%) | 5/7 (71.4%) |
| U1 | ChatGPT | 7 | 3 | 0/7 (0.0%) | 5/7 (71.4%) | 2/7 (28.6%) |
| U2 | CurriculumAI | 7 | 4 | 0/7 (0.0%) | 0/7 (0.0%) | 7/7 (100.0%) |
| U2 | ChatGPT | 7 | 4 | 0/7 (0.0%) | 2/7 (28.6%) | 5/7 (71.4%) |

For useful explanations (U1), agreement was 5/7 (71.4%) for CurriculumAI and 2/7 (28.6%) for ChatGPT, a difference of three respondents. For task help (U2), agreement was 7/7 (100.0%) and 5/7 (71.4%), respectively. Both U2 medians were 4, and neither usefulness item attracted disagreement for either system.

## 5 Perceived trustworthiness

T1: I felt confident in the reliability of this system's answers.

T2: This system provided enough information for me to judge whether its answers could be trusted.

| Item | System | n | Median | Disagreement 1–2 | Neutral 3 | Agreement 4–5 |
|---|---|---|---|---|---|---|
| T1 | CurriculumAI | 7 | 4 | 0/7 (0.0%) | 0/7 (0.0%) | 7/7 (100.0%) |
| T1 | ChatGPT | 7 | 3 | 1/7 (14.3%) | 4/7 (57.1%) | 2/7 (28.6%) |
| T2 | CurriculumAI | 7 | 4 | 1/7 (14.3%) | 1/7 (14.3%) | 5/7 (71.4%) |
| T2 | ChatGPT | 7 | 3 | 3/7 (42.9%) | 1/7 (14.3%) | 3/7 (42.9%) |

All seven respondents agreed with CurriculumAI’s reliability statement (T1), compared with two for ChatGPT. On information sufficient to judge trust (T2), five agreed for CurriculumAI and three for ChatGPT. CurriculumAI also received one neutral and one disagreement rating on T2; ChatGPT received one neutral and three disagreement ratings. These are perceptions rather than verified measures of technical accuracy.

## 6 Open ended feedback themes

Four CurriculumAI comments and three ChatGPT comments contained interpretable substantive feedback. N/A, no-comment entries, uninterpretable text and punctuation-only responses were retained in the coding ledger without substantive themes. Descriptive codes were grouped into the themes below. Each respondent counts at most once within a theme; themes may overlap. Counts use all seven respondents per system. No independent second coding or inter-rater reliability is claimed.

| System | Theme | n / 7 | Explanation | Representative original quotation |
|---|---|---|---|---|

| System | Theme | n / 7 | Explanation | Representative original quotation |
|---|---|---|---|---|
| CurriculumAI | Perceived reliability and trust | 3/7 (42.9%) | Comments describing answers as reliable or trustworthy, including a favourable comparison with ChatGPT. | “CurriculumAI was more reliable than chatgpt” (R01; source W2) |
| CurriculumAI | Conversation continuity without sign-up | 1/7 (14.3%) | A request to access earlier conversations and resume them without signing up. | “I would still be willing to see the previous chat history from without sign up feature so that at some point i coult restart the conversation where i stopped earlier” (R07; source W8) |
| ChatGPT | Perceived accuracy and reliability concerns | 2/7 (28.6%) | Comments expressing that answers could be inaccurate, wrong or unreliable. | “Sometimes it gives wrong answers” (R05; source AL6) |
| ChatGPT | Perceived paid/unpaid answer variation | 1/7 (14.3%) | A reported difference in answers between paid and unpaid access. | “Paid tiers doesn’t give the same answer as the unpaid ones” (R04; source AL5) |

The reliability/trust theme for CurriculumAI was expressed by R01, R03 and R05. R07’s comment is a request for conversation continuity without sign-up, not an independently established defect. ChatGPT accuracy/reliability concerns were expressed by R01 and R05. The paid/unpaid remark does not establish which access was preferable or which configurations were used.

No substantive comment explicitly mentioned citations. A citation-specific explanation for the trust ratings therefore cannot be established from these written responses. Quotations retain original wording, including typographical errors; only outer whitespace is omitted.

## 7 Integrated interpretation

The six complete within-participant comparisons consistently favoured CurriculumAI on perceived usability. Agreement was also higher for CurriculumAI on all four additional items. The written reliability and trust comments are consistent with that descriptive pattern, although the small number of substantive comments limits thematic breadth.

Positive ChatGPT findings remain visible: five respondents agreed that it helped with the tasks, its task-help median matched CurriculumAI at 4, and one available ChatGPT SUS score was 90.00. CurriculumAI's T2 results also contained a neutral and a disagreement response. The pattern supports a cautious account of perceived usability, usefulness and trustworthiness under RQ4; it does not demonstrate factual accuracy or a population-level ranking.

## 8 Methodological limitations

- Seven respondents provide a small descriptive sample. One respondent represents 14.3% of the full sample and 16.7% of the six-pair SUS subset; small count differences should not be overstated.
- One missing SUS response reduces complete CurriculumAI scores and matched pairs to six. Available-data and matched-pair populations must remain separate. No rating or score was imputed.
- Likert ratings are ordinal and are reported using distributions, grouped percentages and medians. Usefulness and trustworthiness remain separate constructs.
- The questionnaire captures perceptions and does not independently assess factual correctness, identify causal mechanisms or support generalisation to a wider student population.
- Qualitative evidence comprises only four and three substantive comments, including very brief statements. Coding is interpretive and has not been independently replicated.
- Assigned-first counts are context only. No order-effect analysis, inferential significance testing, normative SUS grading or model-configuration comparison was undertaken.

## Calculation and source checks

All observed rating prefixes were checked against their labels. SUS scores were independently calculated with two equivalent arithmetic formulations and checked in Microsoft Excel. Item frequencies, percentage denominators, matched membership and theme counts were verified. Every representative quotation was checked against its source cell. A disposable recalculation check changed one answered rating and one coding flag; those changes were discarded. The missing P8 response was never filled, and the supplied source file remains unchanged.